Freight rate prediction

Model, validation and December forecast

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October MAE	October MAPE	Baseline MAE	MAE reduction
$99.29	**4.66%**	$231.30	**57.1%**

Evaluation design

The final inputs cover November-December 2025, after the January-October labeled data. I therefore validated forward in time, keeping every date in one partition. The revision selects by mean dollar MAE across two pre-October forecast windows.

Purpose	Fit period / rows	Evaluate or predict / rows
Earlier comparison	Jan-May / 24,023	Jun-Jul / 9,695
Later comparison	Jan-Jul / 33,718	Aug-Sep / 9,429
Fixed-model check	Jan-Sep / 43,147	October / 4,853
Two-month stress check	Jan-Aug / 38,477	October / 4,853
Final fit	Jan-Oct / 48,000	Nov-Dec / 12,000; fixed chart / 31

Candidate comparison

Model / dollar MAE	Jun-Jul	Aug-Sep	Mean
Original CatBoost	129.21	90.25	109.73
CatBoost, depth 4 / faster learning	130.75	89.88	110.32
CatBoost, depth 5 / faster learning	131.83	89.73	110.78
Smooth baseline + depth-4 trees (selected)	121.77	86.97	104.37
Smooth baseline + depth-6 trees	129.27	89.18	109.23

Each preprocessing fit uses only that window's training rows. The chosen revision gives October RMSE $645.89 and R2 0.821. The day-block bootstrap 95% MAE interval is $84.14-$115.09 (2,000 resamples; conditional on this month). October was inspected in the original assessment; revision selection uses only earlier months. This is a reused chronological check, not a fresh blind test.

Error behavior

The original model's October MAE was $100.26; the revision reduces it by 0.97%. The paired day-bootstrap 95% interval for the MAE improvement is $0.07-$1.93. 98.33% of October predictions are within 10% of the actual price. This is a tolerance-based success rate, not classification accuracy.

October median absolute error is $25.55 and its 95th percentile is $138.91. The largest 1% of absolute residuals account for 97.4% of squared error. No evaluation rows were removed. The fixed two-month horizon check gives $97.24 MAE and 4.55% MAPE; it did not change the model.

Data quality and model

Input checks, missing weights and model features

Issue	Labeled data	Final prediction data	Treatment
Missing weight	300	165	Training-fold equipment median; global fallback
Negative weight	292	145	Treat as invalid; same imputation; add flag
Missing market index	374	249	Exclude from selected model
Duplicate IDs / rows	0 / 0	0 / 0	Fail on duplicate prediction IDs
Long target tail	Max $25,533	No labels supplied	Retain all labeled rows; robust MAE loss

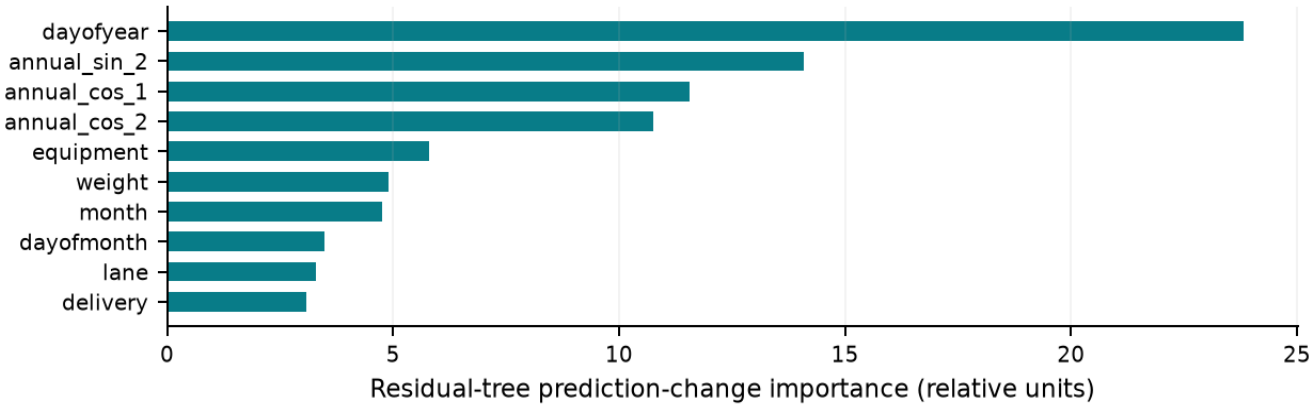
Why exclude the auxiliary signals?

The quote-derived baseline has $317.26 MAE in August but $71.75 in September. That changing reliability suggests a proxy or availability risk; it does not prove leakage. The matched CatBoost ablations confirm that including the signals hurts the two-month development result. Both are also absent from the fixed December table. The final model uses the same six-input contract for every prediction.

Estimator and feature engineering

A regularized Huber additive model first learns smooth distance/weight effects and city, equipment and weekday effects, without annual harmonics. CatBoost then predicts its signed remaining error per mile. Distance-proportional weights align this loss with dollar MAE. The correction is added to the baseline and the total is floored at $1. Tree settings: 2,400 iterations, depth 4, learning rate 0.045, L2 8, seed 42, four CPU threads.

Features include directional city/lane categories, equipment, cleaned weight and its flag, distance/log/inverse distance, weekday/weekend, day, month, day of year and two annual sine/cosine harmonics. City coordinates are derived from fitting-fold lookups. No load ID, manually computed target aggregate, validation label or auxiliary signal enters the feature table. CatBoost learns its categorical statistics inside each training fit. The tree count is fixed; October is not used for early stopping.

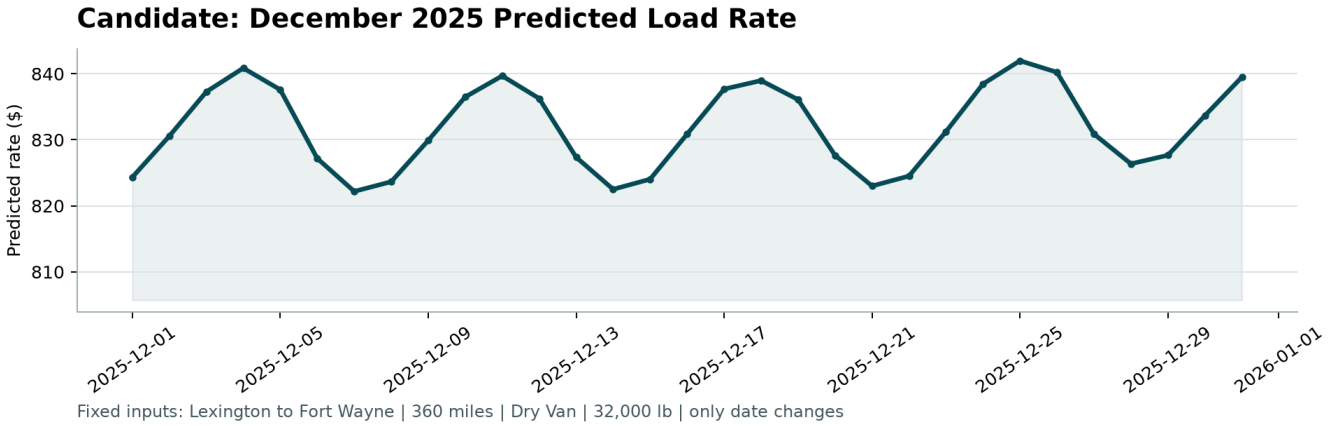


Importance describes only the residual trees, not the additive component or causal effects. Correlated calendar/location features can share importance.

Fixed December forecast

Lexington to Fort Wayne | 360 miles | Dry Van | 32,000 lb

This is the unmodified image produced by the supplied **score.py**. Only the date varies across the 31 rows. The same final model, fitted on all 48,000 labeled loads, produced this curve and the 12,000 load-level predictions.



Daily predicted rates range from $822.18 to $841.97. The scorer validates both CSVs and generates this chart; Spotter calculates hidden-label accuracy after submission.

Run instructions and output checks

The repository includes source, input setup instructions, the original scorer, requirements and an exact environment lock. Raw assessment datasets stay local. **tune.py** recreates revision selection; **train.py** evaluates the fixed model, saves diagnostics and refits; **predict.py** matches predictions to template IDs. It preserves December input columns and validates one-to-one ID coverage and finite positive rates. Both CSV outputs are written to cents. Portable contract and model-export tests check the workflow.

Run after placing the supplied inputs and installing requirements.txt
python run.py (train, validate CSVs, build PDF)
python -m unittest discover -s tests -v

Limits and next steps

Only ten labeled months from one year are available. Seasonal features cannot establish December holiday peaks or future market shocks. The chart has no observed December prices for this fixed lane and should not be interpreted as validated daily seasonality. Repeat cities and lanes are allowed across time splits, matching the forecasting use case. About 12.2% of final directional lanes were unseen in training; unseen-city or new-year performance was not validated. Monitor future dollar error and drift once new labels arrive.